

Josh O'Connor

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Education

University of Kansas, BS in Mathematics Aug 2023 – May 2027

- GPA: 3.77/4.0
- **Advanced Coursework:** Real Analysis, Differential Geometry, Linear Algebra, Modern Algebra, Enumerative Combinatorics, Graph Theory, Measure Theory, Algebraic Combinatorics

Research Experience

McNair Summer Research Internship, University of Kansas May 2025 – Aug 2025

- Supervisor: Reuven Hodges.
- Studied when posets of southeast Kohnert diagrams are ranked and bounded, and when their associated polynomials are monomial-multiplicity free.
- Wrote a Python script to generate all valid southeast $n \times m$ Kohnert diagrams, as well as generate their poset and polynomial. Once we have the poset and polynomial, we are able to test conjectures for boundedness, rankedness, and monomial-multiplicity freeness. This script also wrote and compiled LaTeX to render an initial diagrams poset.

SURIEM REU / SROP Fellow, Michigan State University May 2026 – July 2026

- Supervisor: Matthew Harper. Studying combinatorial approaches to quantum $\mathfrak{sl}_3$ knot invariants.
- Recruited into the NSF-funded SURIE REU through SROP; selected without a separate application.
- The project aimed to find explicit formulas for the 4th and 5th coefficients of the quantum $\mathfrak{sl}_3$ knot invariant.

Senior Honors Thesis, University of Kansas Fall 2026 – Present

- Advisor: Jeremy Martin. Investigating expansions of the chromatic symmetric function into the star basis, with connections to the Stanley–Stembridge conjecture and recent work of Orellana.

Work Experience

Software Engineer Intern, ScriptPro – Mission, KS May 2024 – Aug 2024

- Delivered five backlog items for an internal tool used by 50 business team members to manage 10 client servers.
- Designed and implemented a profiles feature allowing users to batch actions across multiple servers simultaneously, reducing repetitive workflows.
- Refactored search functionality and modernized the UI to improve usability.

Mathematics Instructor & Tutor, University of Kansas Jan 2024 – Present

- Tutored and assisted in college algebra, remedial math, and Calculus I across multiple programs (KAP, SI, and Math Department).
- Led weekly Supplemental Instruction sessions for Calculus I, creating original problem sets each week.
- Graded coursework for Math 500 (Analysis) and Math 558 (Algebra) at the undergraduate level.
- Promoted to supervisory role in KAP; managed help room staffing, shift rotations, and exam grading for other employees.

Honors and Awards

Edith S. Kornbleet Math Dept. Scholarship Spring 2025

- Awarded \$1000 through math department merit scholarship.

Undergraduate Research Award Summer 2025

- Awarded \$1000 to study the boundedness and rankedness of Kohnert posets.

Wealthy Babcock Math Dept. Scholarship Spring 2026

- Awarded \$1000 through math department merit scholarship.

Dean's List

- Made top honor roll in FA23, SP24, FA24, SP25 for cumulative GPA of 3.75 or higher.

Presentations

O'Connor, J. (2025, July). Kohnert Posets and Polynomials of Southeast Diagrams. Delivered oral presentation at McNair Scholars Research Symposium, University of Kansas, Lawrence, KS.

O'Connor, J. (2025, June). College Student Panel. Invited panelist at TRIO Upward Bound, University of Kansas, Lawrence, KS.

O'Connor, J. (2026, March). Kohnert Posets and Polynomials of Southeast Diagrams. Delivered poster presentation at McNair Scholars Conference, University of Maryland College Park.

Organizations

KU Math Club (President and Co-founder)

- Founded the KU Math Club in SP24
- Organize research talks, professional development workshops, and activities for KU students interested in mathematics.

McNair Scholar's Program

- Accepted into a cohort of 11 students, all dedicated to pursuing a research career in our respective fields
- Attend monthly professional development seminars, as well as various research and scholarly activities.
- The mission of the McNair Scholar's program is to amplify excellence and advance academic and research fields by preparing students for the Ph.D.

KU Esports (Co-Captain)

- Competed weekly against regional schools
- Organized matches with said schools
- Made playoff runs for both seasons played

Skills

Programming: Python, SageMath, $\text{\LaTeX}$, Git

Mathematics Software: SageMath (polynomial rings, symmetric functions, poset generation), TikZ (mathematical diagrams and commutative diagrams)

Non-Academic Pursuits

Competitive Esports

Counter-Strike

- ESEA Main for 3 seasons (semi-professional league)
- Ranked in the top 1000 players in North America (Top 0.1%)